EXPERIMENT 4

IMPLEMENTATION OF RING TOPOLOGY USING PACKET TRACER

Aim

To implement a ring topology using Packet Tracer and to transmit data between the devices connected in the ring topology.

Software Used

Cisco Packet Tracer (end devices, switches, connecting cables)

Procedure

Step 1: Start Cisco Packet Tracer.

Step 2: Click **End Devices**, select the Generic PC and click in the workspace to place it. Place four PCs in the same way.

Step 3: Click **Switches** and place four switches in the workspace.

Step 4: Click **Connections**, connect each PC to one switch, and then connect the four switches to one another in a closed loop (Switch0 - Switch2 - Switch3 - Switch1 - Switch0) to form the ring.

Step 5: Wait until the amber link lights on the switch ports turn green. This happens after the Spanning Tree Protocol (STP) process finishes, and the ports can then forward frames.

Step 6: Click PC0, open the **Config** tab, select FastEthernet0 and type the IP address. The subnet mask is generated automatically. Do the same for every PC with a different IP address.

Step 7: To confirm data transfer, click a PC, open **Desktop** and then **Command Prompt**, and ping the IP address of another PC. The ping statistics are displayed.

Execution Screenshot

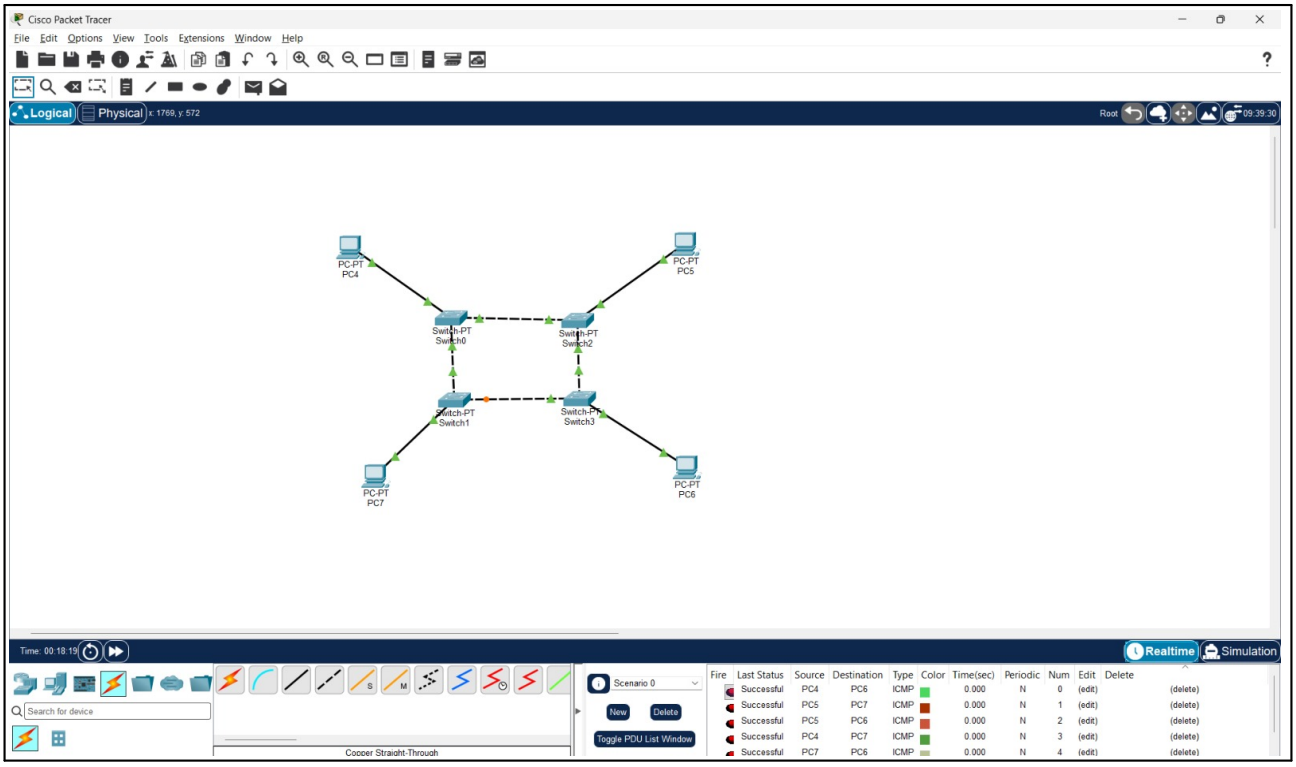


Fig: Ring topology with successful ICMP ping between the PCs

Result

Thus the ring topology is implemented with the Packet Tracer simulation tool.